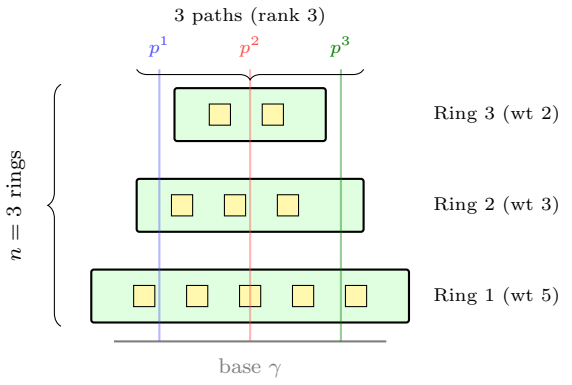
A CPP with max entry $n = 3$



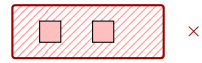
$|\nu_{\text{full}}| = 10 + |\gamma|$

two types of compression

The triple (S, U, ν)

$S = \{3\}$

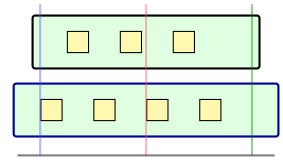
level 3 **crushed** (ring destroyed)



$\sigma(S) = 3$, sign $(-1)^1$

from q-binomial

ν: **max entry** $m = 2$



from GK_c, *sign* $+1$

$U = \{1\}$

ring 1 **compressed** (weight extracted)



$\sigma(U) = 1$, sign $(-1)^1$

from $(q)_m$

Total weight: $\underbrace{\sigma(S)}_{3} + \underbrace{\sigma(U)}_{1} + \underbrace{|\nu|}_{7+|\gamma|} = 11 + |\gamma|$

Sign: $(-1)^{|S|+|U|} = (-1)^{1+1} = +1$

Key distinction:

S: ring *destroyed* (all face labels lost)

U: ring *preserved* (some weight extracted)